X-ray Cavities Honours Project: Stewart Environment Gate Status

Supervisor update for 2026-05-06 meeting

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Executive Summary

Post-meeting correction, `2026-05-07`: this report was prepared before the 2026-05-06 supervisor meeting. The environment-gate results remain valid, but the follow-on plan has changed. The nominal first science run is now particles-enabled `C-HI` to about `50 Myr`, and `JET_END_TIME` / `JET_ACTIVE_TIME` are Myr values in the user parameter block, not PLUTO code-time values.

The main science asset blocker has been cleared. Patrick supplied the Stewart environment archive and GRAiLE/XRAiSE bundle, and Stas clarified the key science choice: production science runs should use the radially averaged tabulated environments, not the full 3D grid environments, because the full 3D profiles introduce too much spurious motion for reliable energetics.

We extracted the two selected tabulated Stewart environments, staged them to a resized Nectar VM, and ran no-jet 3D Cartesian stability gates to `300 Myr`. Both the cluster and group environments passed the existing `jet-setup` stability verifier at smoke scale.

The project is now ready for the corrected next science gate: a production-like, particles-enabled `C-HI` high-power cluster control, preceded by a `kunanyi` no-jet environment/MPI/HDF5 validation at closer-to-production scale.

What Is Now Confirmed

Environment Selection

Case role	Case ID	Stewart environment	Selected file
high-power cluster validation	`C-HI`	cluster `002_003`	`tabulated_cosmo_002003.csv`
low-power cluster science anchor	`C-LO`	cluster `002_003`	`tabulated_cosmo_002003.csv`
low-power group comparison	`G-LO`	group `002_031`	`tabulated_cosmo_002031.csv`

The full 3D environment cubes remain useful reference assets, but are not the default science inputs for the honours energetics path.

Frozen Science Defaults

Parameter	Current default
Physics	HD
Geometry	3D Cartesian
Jet speed	`0.1 c`
Opening angle	`15 deg`
First duration target	`50 Myr`
Early snapshots	about every `1 Myr`; inspect first `3-5`
Views	`0`, `45`, `90 deg`
First validation case	`C-HI`, `JET_PWR = 1e45 erg/s`
Particle staging	particles enabled for nominal first science run

Important unit correction after the meeting: PLUTO core controls use code-time values, but user-defined jet-duty-cycle parameters are physical Myr values. A `50 Myr` first target is rendered as:

```
tstop = 15330.06968940
JET_END_TIME = 50.0
JET_ACTIVE_TIME = 50.0
1 Myr output cadence = 306.60139379
```

Nectar Environment Gate

The MacBook remained the control plane. `nectar-vm` was resized to `c3.3xlarge` for the gate tests, then parked back to `c3.xsmall` afterwards.

Each gate used:

- HD, 3D Cartesian, vector body force
- no jet: `JET_END_TIME = 0`, `JET_ACTIVE_TIME = 0`
- no particles: `PARTICLES = NO`
- uniform `50 x 50 x 50` grid over `[-100, 100] kpc`
- outputs near `0`, `100`, `200`, and `300 Myr`

Results

Environment	Runtime	Max central density drift	Max central pressure drift	Max Mach	Max velocity	Gate
cluster 002_003	39 s	0.136%	0.226%	0.0242	10.6 km/s	pass
group 002_031	30 s	0.284%	0.473%	0.0330	8.07 km/s	pass

The result is not a final production-resolution validation, but it is a clean smoke-scale gate: both tabulated environments load correctly, remain quiet over the full `300 Myr` duration, and produce standard `grid.out`, `dbl.out`, and `data.*.dbl` outputs that load through the current Python analysis environment.

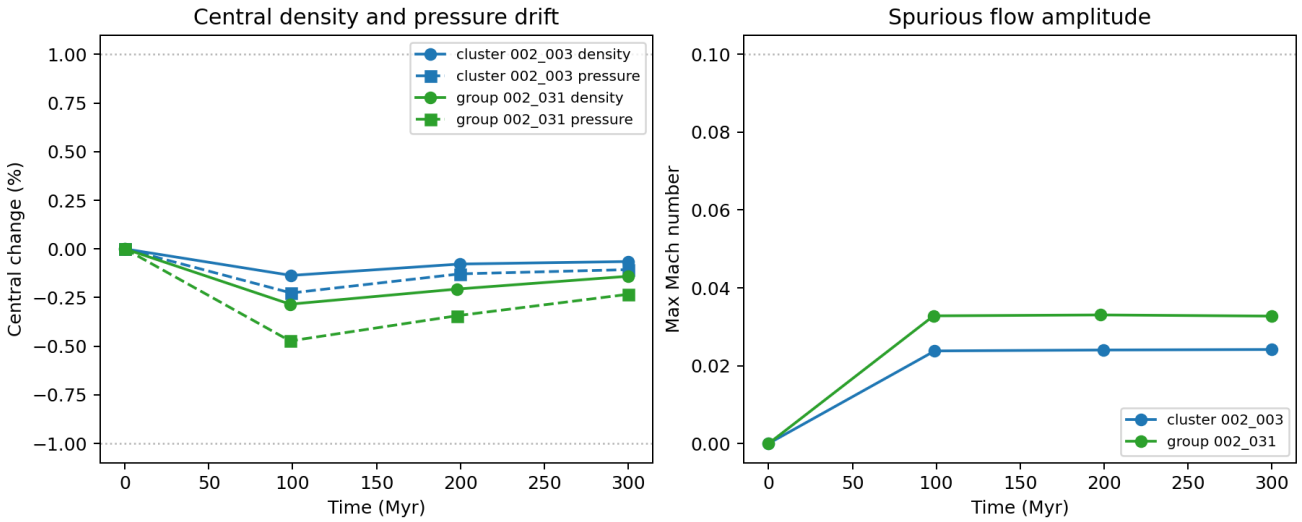


Figure 1: Stability gate time series

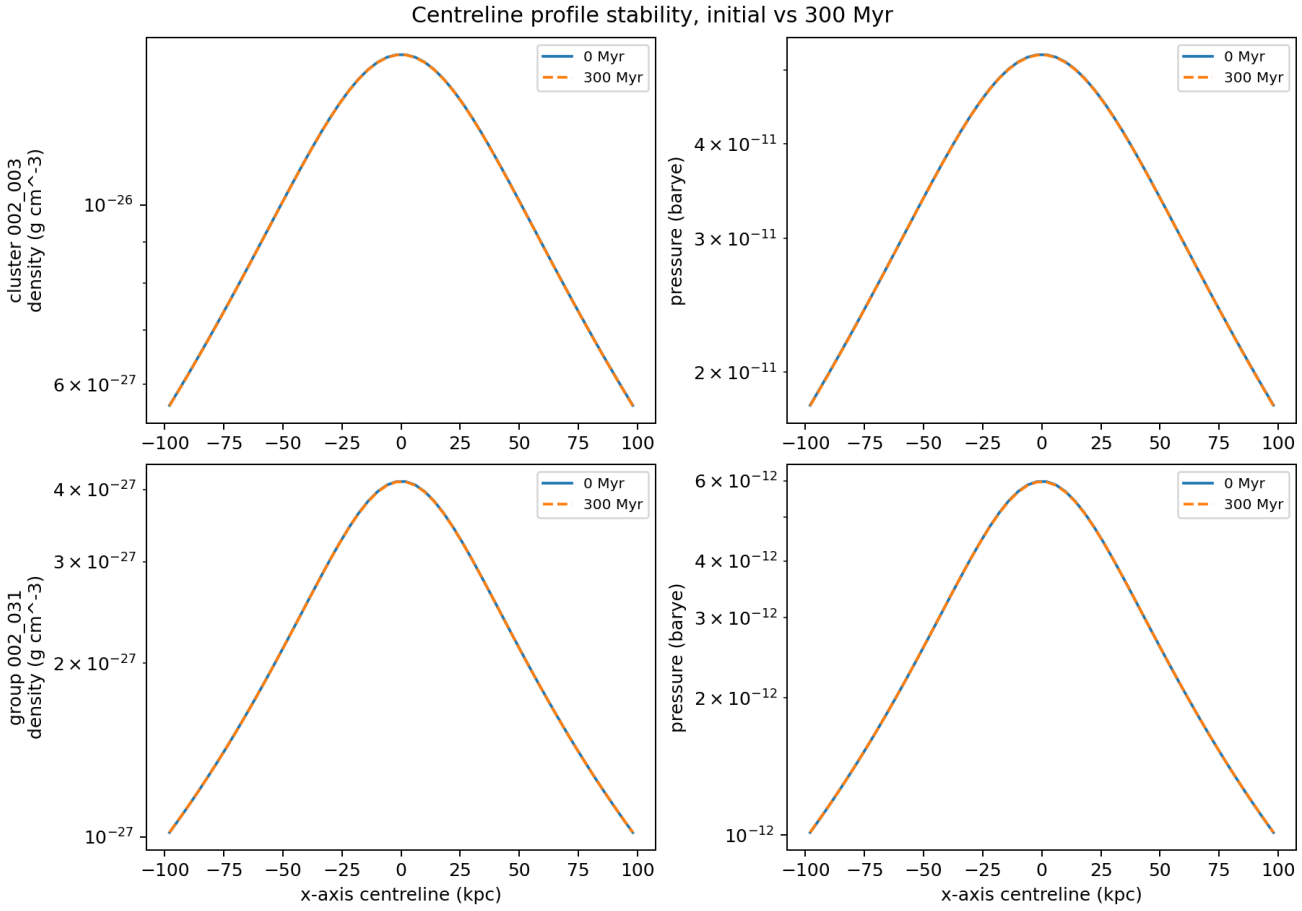


Figure 2: Initial versus 300 Myr centreline profiles

Technical Work Completed

New and Updated Project Files

- `docs/science/stewart_environment_gate_2026-05-05.md`: detailed gate report.
- `docs/science/assets/stewart_environment_gate_2026-05-05/`: metrics and plots.
- `scripts/science/render_stewart_environment_gate_tests.py`: renders `jet-setup`-compatible no-jet stability tests from the Stewart tabulated environments.
- `scripts/science/plot_stewart_environment_gate.py`: regenerates the summary metrics and plots from gate outputs.
- `scripts/science/render_three_case_configs.py`: corrected after the meeting so PLUTO core times are code-time values while jet active/end times remain Myr values.
- `docs/science/three_case_spec.md`: updated to record the tabulated environment decision, the green Nectar gate, and the code-time convention.

Operational Fixes

The current `jet-setup` source template keeps `PARTICLES_LP` enabled in `definitions.h`. For a true no-particle environment gate, the gate renderer now emits a hydro-only `definitions.h` and a `local_make` that includes the UTAS environment and jet user objects without including `Particles/makefile`.

Nectar SSH was also made more stable after intermittent port 22 connection refusals by disabling socket activation and enabling persistent `ssh.service`. After the gate, `nectar-vm` was parked back to `c3.xsmall` and verified by SSH as `1` vCPU and about `1.9 GiB` RAM.

Current Status

The infrastructure path is green:

- MacBook can orchestrate Nectar.
- Nectar can resize up for rehearsals and park down afterwards.
- `kunanyi` has already passed PBS, HDF5, PLUTO, `jet-setup`, email, and multi-node MPI smoke tests.

The science-input path is now green for the first rehearsal:

- Stewart tabulated cluster and group files are identified.
- Both selected files pass no-jet stability smoke tests.
- The GRAiLE/XRAiSE bundle is present locally, but still needs path and dependency porting before it is a repo-native X-ray pipeline.

Immediate Forward Plan

Gate 1: Production-like Kunanyi Environment Check

Repeat the no-jet Stewart tabulated cluster check on `kunanyi`, closer to the production setup:

- at least `1 cell/kpc`, preferably `2 cells/kpc` where feasible
- no jet
- MPI through PBS
- HDF5 output enabled
- available `kunanyi` modules first

Pass criteria:

- compile and run on a compute node
- HDF5 outputs written and readable
- no obvious hydrostatic drift
- no private MPI/HDF5 build needed

Gate 2: Particles-enabled `C-HI` Symmetric Control

Run the first high-power cluster case with particles enabled:

- case: `C-HI`
- environment: cluster `002_003` tabulated profile, centered
- jet power: `1e45 erg/s`, one-sided
- jet speed: `0.1 c`
- opening angle: `15 deg`
- duration target: about `50 Myr`
- early outputs: about `1 Myr` cadence

Pass criteria:

- first `3-5` snapshots are stable and diagnostically sensible
- clear lobe/cavity morphology
- particle output files are produced and load
- no output, restart, or unit surprises

Gate 3: First Observable/Truth Comparison

Build the first aligned comparison package for one `C-HI` snapshot/view:

- Level 1 truth: injected energy, with one-sided and bipolar conventions both reported until supervisors choose the convention.
- Level 2 truth: tracer-defined lobe volume and energy with tracer threshold `5e-3`, plus sensitivity checks at `1e-3` and `1e-2`.
- X-ray path: port GRAiLE/XRAiSE scripts rather than inventing a substitute.
- Cavity protocol: ellipsoidal cavity boundary, local ambient pressure, `pV` and `4pV`, and sound/buoyancy/refill age estimates.

Gate 4: `kunanyi` Intermediate Rehearsal

Use the production PBS template for the intermediate `kunanyi` rehearsal:

```
select=N:ncpus=28:mpiprocs=28
rank count from PBS_NODEFILE
#PBS -M barry.brook@utas.edu.au
#PBS -m ae
```

The production launch order remains:

1. `C-HI`
2. `C-LO`
3. `G-LO`

Supervisor Checkpoints

The useful discussion points for the 2026-05-06 meeting are:

1. Confirm that the tabulated Stewart files are now the agreed science inputs.
2. Confirm whether `JET_INITIAL_RADIUS` and `JET_INJECTION_HEIGHT` should stay near the historical Stewart values or be adjusted for the honours grids.
3. Confirm the preferred first snapshot for the aligned X-ray/radio/truth smoke comparison.
4. Confirm the production grid patch structure and storage cadence.

Bottom Line

We are no longer blocked on the Stewart environment files. The next meaningful science task is an auditable, particles-enabled `C-HI` package and production-like `kunanyi` validation path, with early diagnostics as the first science-facing deliverable.